

GHOST PREDATOR — Book-Depth Gate (MIN_BOOK_DEPTH)

Skip windows where the order book is too thin to trust · build guide for the team · June 11, 2026

The Problem: a thin book is a lie detector failure

The whole favorite-snipe premise assumes the ask price is INFORMATION — that 74¢ means "the market is 74% sure." That's only true when real money is resting in the book. In dead hours, the book can hold \$0–\$30 total: the "74¢ favorite" is one stale order nobody cancelled, not consensus. Two live case studies (June 11, ~4am UTC): an ETH window with **\$26 total book volume** sold us Down at 74¢ while the official (Chainlink) outcome was Up wire-to-wire; the next window had a **\$0-volume book** and a \$13 in-band ask shelf. Both lost. Dead books are also where sub-floor fills (0.44 / 0.352 against a 0.50 floor) and FOK kills cluster — quotes are stale, races are worst, and in flat windows the oracle's own \$1–2 noise decides the coin flip.

The Signal: in-band resting depth at fire time

Measure the dollars actually resting on the ask side at or below your MAX_ASK cap, at the moment you'd fire:

```
cum_usdc = sum(price * size for price, size in book[token].items() if price <= MAX_ASK)
if MIN_BOOK_DEPTH > 0 and cum_usdc < MIN_BOOK_DEPTH:
    skip this window # log it once: '[skip-dead] ... in-band book $X < $100; not fired'
```

This is NOT the fill check (your \$10–\$22.50 stake only needs that much depth to fill). It is a market-quality bar: enough independent money behind the quote that the price means something. It scales with the market, not your bankroll — \$100 is the same bar for a \$250 account or a \$25k one.

Evidence (762 resolved live trades, May 28 – Jun 11)

In-band depth at fire	N	WR	Net P&L	\$/trade
< \$50	223	71.7%	–\$91.84	–\$0.41
\$50–100	82	73.2%	–\$55.56	–\$0.68
\$100–300	159	69.8%	–\$88.77	–\$0.56
\$300–1000	178	72.5%	+\$57.52	+\$0.32
\$1000+	120	66.7%	–\$19.79	–\$0.16

Rule	Trades kept	Net P&L
baseline — no gate	762 (100%)	–\$198.44
skip depth < \$50	539 (71%)	–\$106.60
skip depth < \$100 ← deployed	457 (60%)	–\$51.04
skip depth < \$200	359 (47%)	+\$12.59
skip depth < \$300	298 (39%)	+\$37.73

Honest read: the trend is strong (~75% of the bleed lives below \$100) but the middle bands aren't perfectly monotonic and this overlaps heavily with overnight hours — depth is the cleaner causal variable than clock time, but treat the exact threshold as directional, not gospel. The gate only REMOVES trades; it cannot add risk.

Build It (generic — adapt names to your build)

1. **Config knob:** `MIN_BOOK_DEPTH = float(os.getenv("MIN_BOOK_DEPTH", "0"))` (0 = off, so the default changes nothing).
2. **Placement:** in your snipe/fire loop, AFTER you've picked the candidate token and BEFORE any order is built — right where you already read the order book for the fill check. You need the full in-band ask ladder (WSS book snapshot), not just the best ask.
3. **Gate:** compute cum in-band depth and continue if below threshold. Log once per token per window (dedup set) so the log stays readable.
4. **Do NOT** apply it to exits/redeems — entries only.

```
# inside the per-market fire evaluation, before sizing/signing:
cum_usdc = round(sum(p * s for p, s in books.get(tok, {}).items() if p <= MAX_ASK), 2)
if MIN_BOOK_DEPTH > 0 and cum_usdc < MIN_BOOK_DEPTH:
    if tok not in skip_logged:
        log(f"[skip-dead] {coin} {side} {dur} - in-band book ${cum_usdc:.0f} "
            f"< ${MIN_BOOK_DEPTH:.0f}; ask price not trustworthy, not fired")
        skip_logged.add(tok)
    continue
```

Tune It On Your Own Data

If your bot records in-band depth per trade (ours stores it as `cum_usdc` on each position), backtest your own threshold before picking one:

```
-- per-band: where does YOUR bleed live?
SELECT CASE WHEN cum_usdc<50 THEN '<50' WHEN cum_usdc<100 THEN '50-100'
           WHEN cum_usdc<300 THEN '100-300' WHEN cum_usdc<1000 THEN '300-1000'
           ELSE '1000+' END AS band,
       COUNT(*) n, ROUND(AVG(status='won')*100,1) wr, ROUND(SUM(pnl),2) net
FROM positions WHERE status IN('won','lost') GROUP BY band;

-- backtest a threshold K:
SELECT COUNT(*), ROUND(SUM(pnl),2) FROM positions
WHERE status IN('won','lost') AND cum_usdc >= 100; -- try 50/100/200/300
```

Start at \$100 (balanced) unless your data clearly says otherwise. Expect overnight volume to drop sharply — that is the gate working. Watch for [skip-dead] lines as proof of life. Revisit the threshold after ~2 weeks of post-gate data.